



A shock tube and modeling study on the auto-ignition properties of pyrrole in O₂/CO₂ atmosphere at elevated pressures

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ABSTRACT

Understanding the details of pyrrole combustion chemistry in the O₂/CO₂ atmosphere contributes to developing the strategies for nitrogen oxides (NO_x) control during the pressurized oxy-coal combustion (POCC). However, the existing kinetic models for pyrrole oxidation lack multi-dimensional validation, and the ignition delay times (IDTs) of pyrrole in the O₂/CO₂ atmosphere are still scarce. This study measured the IDTs of pyrrole in O₂/CO₂ atmospheres at an elevated pressure of 5.2 bar, temperatures from 1271 to 1645 K, and equivalence ratios $\Phi = 0.5, 1.0$, and 2.0 in a shock tube. The results demonstrate that the pyrrole IDTs decrease with decreasing equivalence ratio and the pyrrole mixtures in O₂/CO₂ atmospheres have longer IDTs than those in O₂/Ar atmospheres. A pyrrole oxidation kinetic model (HUST pyrrole model) has been developed by updating 19 reactions in our previous model (HUST pyridine model). HUST pyrrole model gives a satisfactory prediction of the IDTs of pyrrole in the O₂/Ar atmosphere (measured by MacNamara et al.) and O₂/CO₂ atmosphere (measured in this study) and the profiles of pyrrole, HCN, and NO (measured by Yamamoto et al.). The comparison of the HUST pyrrole model with the Lumbreras model (2001) was conducted by the pyrrole IDTs and profiles of HCN using the sensitivity and flux analysis. The addition of HNCPPROP and modification of R665 ($C_3H_3 + O_2 = CH_2CO + HCO$), R1273 ($AC_3H_4CN + H_2 \Rightarrow ALLYL CN + H$), R1251 ($PYRLNE = ALLYL CN$), R1238 ($PYRROLE + H = PYRLYL + H_2$), and R1240 ($PYRROLE + OH = PYRLYL + H_2O$) in HUST pyrrole model significantly improve the prediction performance for the pyrrole oxidation. The effects of the equivalence ratio and CO₂ on the IDTs for pyrrole oxidation have been analyzed.

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1. Introduction

Oxy-coal combustion technology is considered as a suitable and promising technology for reducing carbon dioxide (CO₂) emissions from power plants [1]. Pressurized oxy-coal combustion (POCC) technology, as an advanced oxy-coal combustion technology, has been gaining a lot of concerns owing to its high efficiency, low emission, compact dimension of the physical furnace and associated equipment [2–5]. In addition to the reduction of CO₂ emissions, the oxy-coal process offers opportunities for new methods of NO_x control [6]. Therefore, the study on nitrogen chemistry during POCC is of the essence for the development of strategies for NO_x control.

During oxy-coal combustion, fuel NO is generally the dominant source of NO because of the absence of N₂. Since nitrogen in coal localizes mainly in pyrrolic and pyridinic aromatic nitrogenous het-

erocycles structures [7,8], pyrrole has been widely used as a representative compound to study the fuel-N conversion and oxidation mechanism [9–19]. In 1991, Mackie et al. [9] investigated the pyrrole pyrolysis in a shock tube at pressures of 7.5–13.5 atm, and temperatures from 1200 to 1700 K and proposed a pyrrole pyrolysis model with 75 reactions, which tolerably fit the mole fractions of the major products of pyrrole pyrolysis. In 1999, Martoprawiro et al. [10] investigated many pyrolysis reactions of pyrrole by ab initio methods and developed a kinetic model for pyrrole pyrolysis including a direct ring-opening reaction of pyrrole ($PYRROLE = HNCPPROP (HNCHCHCCH_2)$) and imine HNCPPROP related reactions ($PYRLNE = HNCPPROP$ and $HNCPPROP = C_3H_4 + HCN$), which was validated by the experimental results from Mackie et al. [9].

In 2000, during the investigation of the pyridine oxidation, Ikeda et al. [20] obtained the rate constants for some pyrrole oxidation reactions in part from the ab initio potential energy surfaces and partly by thermochemical estimation. In 2001, Lumbreras et al. [11] developed a pyrrole oxidation model based on the mechanisms from Ref. [9,10], and [20] (hereafter referred to

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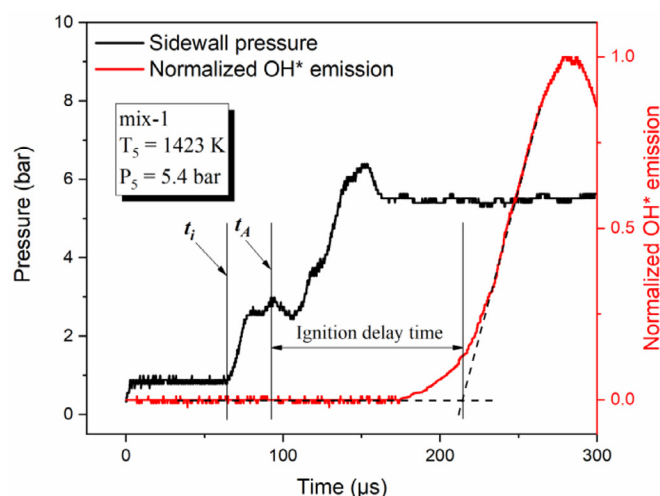


Fig. 1. Typical OH* and pressure profiles for mix-1 at 1423 K and 5.4 bar.

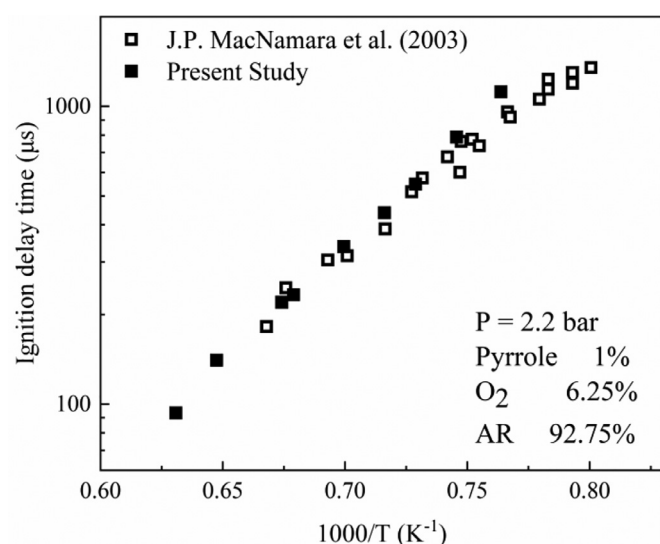


Fig. 2. Comparison of the IDTs of pyrrole measured in the present study to those from MacNamara et al. [12].

as Lumbreras model) but there were evident quantificational disparities between the prediction and experiments for the profiles of CO, CO₂, and HCN measured in a plug flow reactor (PFR) at atmospheric pressure. It is noted that the Lumbreras model excluded the three HNCPPROP related reactions (PYRROLE = HNCPPROP, PYRLNE = HNCPPROP, and HNCPPROP = C₃H₄ + HCN). In 2003, Koger et al. [13] improved the Martoprawiro et al. mechanism [10] by increasing the rate constants of two reactions (PYRROLE = HNCPPROP and HNCPPROP = C₃H₄ + HCN) by a factor of 30 to fit the profiles of pyrrole and HCN during the pyrrole oxidation process. In 2012, based on Ref. [9,11,13], Yamamoto et al. [14] developed a pyrrole oxidation model. The model predictions agree well with the profiles of pyrrole, HCN, and NO obtained in a PFR diluted in N₂ at atmospheric pressure and temperatures of 800–1400 K, but the model was not provided in [14]. However, the existing kinetic models for pyrrole oxidation lack multi-dimensional validation.

The IDTs measured in a shock tube are a crucial parameter for the kinetic model validation. In 2003, MacNamara et al. [12] reported the IDTs of pyrrole using argon as a diluent at pressures of 1.9 to 5.6 bar, equivalence ratios of 0.5–2.0, and temperatures of 1102–1805 K, but none of the existing models can predict well the experimental results. As far as we know, experimental data of

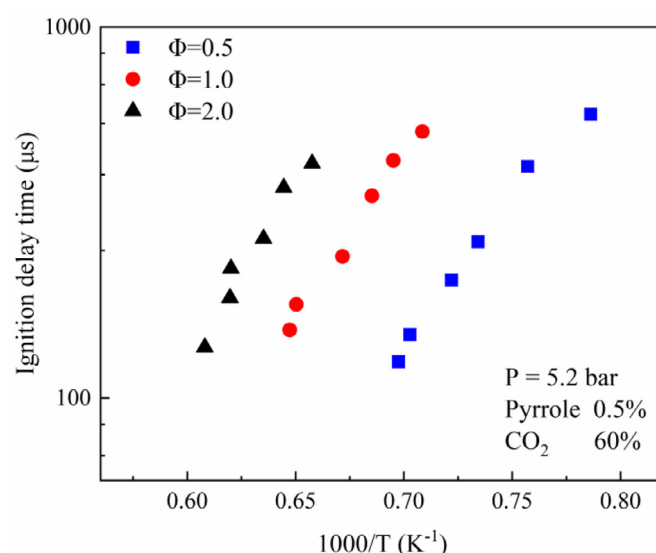


Fig. 3. Measured IDTs for 0.5% pyrrole, at $P = 5.2$ bar, at various equivalence ratios under O₂/CO₂ atmospheres.

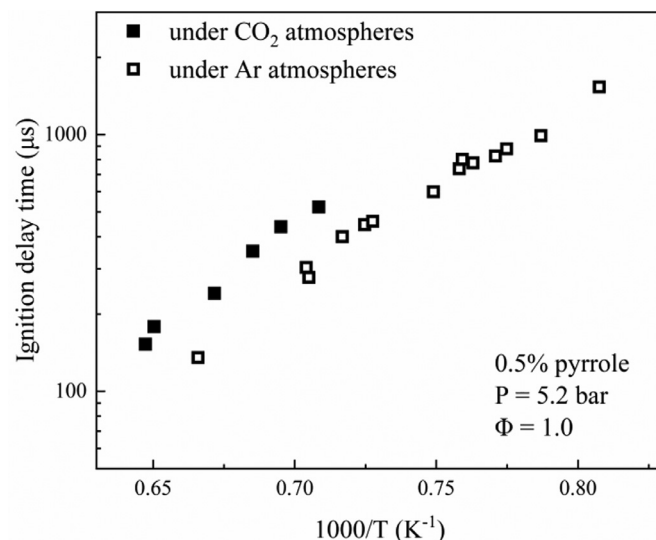


Fig. 4. Comparison between the IDTs in O₂/CO₂ atmospheres and those in O₂/Ar atmospheres for stoichiometric pyrrole mixtures at $P = 5.2$ bar.

the IDTs for pyrrole oxidation under O₂/CO₂ atmospheres are still scarce.

Because the kinetic models for the pyridine oxidation include the pyrrole oxidation sub-mechanism, both of the kinetic models of the pyridine oxidation proposed by Alzueta et al. [21] in 2002 and Wu et al. [22] in 2019 employed the Lumbreras model as the pyrrole sub-mechanism. HUST pyridine model proposed in our previous study [23,24] employed the Lumbreras model with the addition of HNCPPROP related reactions (PYRROLE = HNCPPROP, PYRLNE = HNCPPROP, and HNCPPROP = C₃H₄ + HCN) as the pyrrole sub-mechanism. However, the three pyridine models above fail to satisfactorily predict both the species profiles of pyrrole oxidation from Yamamoto et al. [14] and IDTs of pyrrole oxidation from MacNamara et al. [12].

Obviously, this work aims to study the pyrrole oxidation mechanism in O₂/CO₂ atmospheres. The IDTs of pyrrole under the O₂/CO₂ atmosphere were measured over temperatures from 1271 to 1645 K, at an elevated pressure of 5.2 bar, and equivalence ratios $\Phi = 0.5, 1.0$, and 2.0. Subsequently, a pyrrole oxidation model,

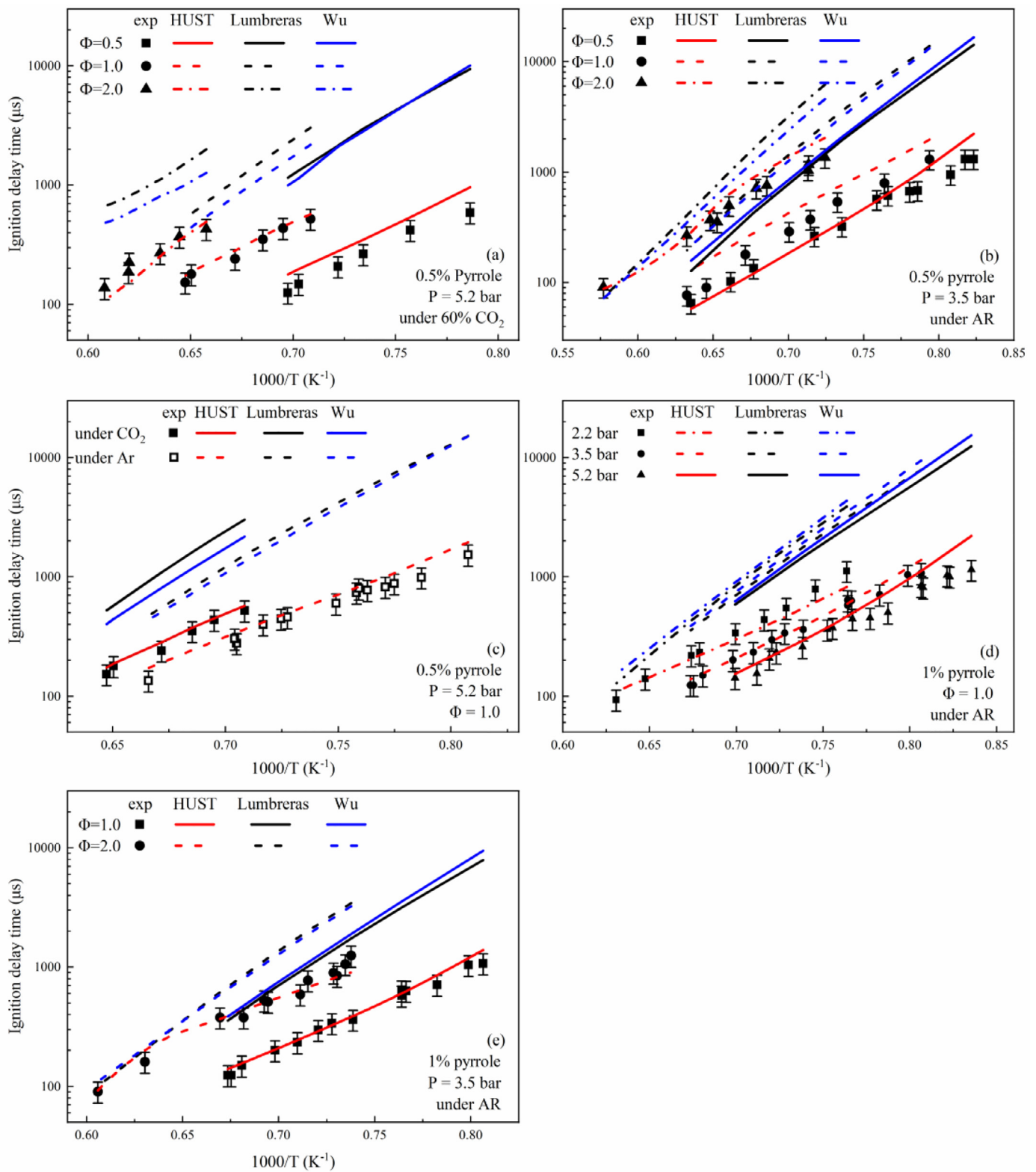


Fig. 5. The IDT measurements for pyrrole/ O_2 mixtures from the present work and MacNamara et al. [12] and simulation results obtained with HUST pyrrole model, Lumbreras model, and Wu model.

named as HUST pyrrole model, was developed on the basis of the HUST pyridine model [24] and validated by a wide range of pyrrole and pyridine experimental data from the literature. The comparison of the HUST pyrrole model with the Lumberras model [21] was conducted with further flux and sensitivity analysis against the IDTs of pyrrole in the O_2/Ar and O_2/CO_2 atmosphere and the mole fractions of HCN during pyrrole oxidation in the O_2/N_2 atmosphere obtained in a PFR. The effects of the equivalence ratio and CO_2 on the IDTs of pyrrole have been analyzed.

2. Experimental methods

In this work, the IDTs for the oxidation of pyrrole under O_2/CO_2 atmospheres were measured in a shock tube. The details of the shock tube used in the present study are described in our previous work [25]. Briefly, the shock tube with an inner diameter of 0.1 m consists of a 4-m driver section and an 8-m driven section. These two sections are separated by two polyester terephthalate diaphragms. Table 1 shows the pyrrole mixture compositions (mole%). Test mixtures were prepared manometrically and allowed to rest for more than 12 h in a 300 L mixing tank. The partial pressure of pyrrole was kept below 50% of its saturation vapor pressure (1.1 kPa at 25 °C) to ensure complete vaporization.

Table 1
Experimental conditions.

Mixture	Pressure (bar)	Φ	% Pyrrole	% O_2	% CO_2	% Ar
1	5.2	0.5	0.5	6.25	60	33.25
2	5.2	1.0	0.5	3.13	60	36.37
3	5.2	2.0	0.5	1.56	60	37.94

Helium (99.999% purity) and nitrogen (99.999% purity) were used as the driver gas. The purities of pyrrole (Aladdin), argon, oxygen, and CO_2 are 99%, 99.999%, 99.999%, and 99.999%, respectively. The driven section was vacuumed down to 0.1 Pa prior to every run.

The temperature (T_5) and pressure (P_5) behind the reflected shock wave were calculated using the Gaseq [26] chemical equilibrium program. The uncertainty for the determination of T_5 , which is the most important uncertainty, was less than 21 K according to the method used in our previous work [25,27]. The total uncertainty in the IDT measurements was less than 20%. The detailed analysis of the uncertainty can be found in the supplemental material.

The IDT is defined as the time interval between the arrival of the reflected shock wave and the ignition point, indicated by the

Table 2
Chemical nomenclature for important species in the process of pyrrole oxidation.

Species Name	Structure	Name in the model
Pyrrole		PYRROLE
Allenic imine		HNCPROP
Pyrrolenine		PYRLNE
Pyrrolyl radical		PYRLYL
Allyl cyanide		ALLYLCN
Crotonitrile		C_3H_5CN
Cyanoallyl radical		AC_3H_4CN
Propyne		C_3H_4-P
Allene		C_3H_4-A
Propargyl radical		C_3H_3

intersection of lines drawn along the steepest rise of OH* signal and the zero baseline, as depicted in Fig. 1. To overcome the difficulty of the arrival time determination of the reflected shock due to shock bifurcation caused by the high concentration of CO₂, an empirical expression proposed by Peterson et al. [28] was used. The arrival time of the reflected shock wave, t_A , can be related to the rise time of the pressure, t_i , via

$$t_A - t_i = \Delta t_{AO} + D/2V_R \quad (1)$$

$$\Delta t_{AO} (\mu s) = 4.6 M_S^{0.66} \gamma_2^{-7.1} \bar{M}^{0.57} \quad (2)$$

where D denotes the diameter of the piezoelectric pressure transducer (Kistler 603B1), M_S denotes the Mach number of the incident shock, $\bar{M}$ is the molecular weight of the mixture, V_R denotes the velocity of the reflected shock wave, γ_2 denotes the specific heat of the upstream mixture.

3. Modeling

Numerical calculations of IDTs applying the adiabatic and constrained volume assumptions were carried out by the closed zero-dimensional module from CHEMKIN-PRO [29]. The calculated IDT is defined as the time interval between zero and the moment of the maximum rate of temperature rise (max dT/dt).

Chemical nomenclature for important species in the process of pyrrole oxidation are reported in Table 2. The HUST pyrrole model was developed based on the sub-model for pyrrole oxidation in the HUST pyridine model [23,24] by updating 19 reactions, as listed in Table 3.

High-level quantum chemistry calculation is a reliable method to obtain rate constants for reactions. Mackie et al. [9] estimated the rate coefficients of the unimolecular H-abstraction reaction of pyrrole, $\text{PYRROLE} = \text{PYRLYL} + \text{H}$ (R1237), in analogy with cyclopentadienyl and suggested that activation energy of R1237 larger than 84 kcal/mol can lead to better performance. The latest Arrhenius parameters of R1237 calculated by Bacskay et al. [30] using an ab initio method were adopted in the HUST pyrrole model. Lumbieras et al. [11] estimated the Arrhenius parameters of the H-abstraction reactions of pyrrole and pyrrolene by OH, $\text{PYRROLE} + \text{OH} = \text{PYRLYL} + \text{H}_2\text{O}$ (R1240) and $\text{PYRLNE} + \text{OH} = \text{PYRLYL} + \text{H}_2\text{O}$ (R1257), by analogy to the hydrogen abstraction reaction of phenol reaction. The Arrhenius parameters of R1240 used in the HUST pyrrole model were sourced from the theoretical calculation results in Ref. [18] in 2020. Similar to Lumbieras et al. [11], the Arrhenius parameters of R1257 are the same as R1240 in the HUST pyrrole model.

In fact, it is very difficult to calculate or measure the rate constants of many pyrrole related reactions due to the complex heterocyclic aromatic structure and experiment limitation. Therefore, the analogy method has been frequently employed to estimate the rate constants of pyrrole related reactions. Mackie et al. [9] estimated the Arrhenius parameters of $\text{PYRROLE} + \text{H} = \text{PYRLYL} + \text{H}_2$ (R1238), $\text{PYRLNE} = \text{ALLYLCN}$ (R1251), and $\text{AC}_3\text{H}_4\text{CN} + \text{H}_2 = \text{ALLYLCN} + \text{H}$ (R1273) using the analogy method. It is noteworthy that the rate constants of R1251 were increased by a factor of 0.88 by Mackie to predict well their experiment. Moreover, the rate constants of $\text{PYRROLE} = \text{HNCPROP}$ (R1531) calculated theoretically by Martoprawiro et al. [10] was increased by a factor of 30 in Ref. [13]. Therefore, the Arrhenius parameters of R1238, R1273, R1251, and R1531 were globally optimized in the present work and the results were shown in Table 3.

The rate constants of $\text{AC}_3\text{H}_4\text{CN} + \text{H} = \text{ALLYLCN}$ (R1284) and $\text{AC}_3\text{H}_4\text{CN} + \text{H} = \text{C}_3\text{H}_5\text{CN}$ (R1285) were sourced from Doughty et al. [31].

Propargyl radical (C_3H_3) is an important resonantly stabilized free radical during the oxidation process of pyrrole [32]. Consequently, the Arrhenius parameters of the reaction between C_3H_3

with oxygen, $\text{C}_3\text{H}_3 + \text{O}_2 = \text{CH}_2\text{CO} + \text{HCO}$ (R665), were sourced from the calculated results by Hahn et al. using RRKM theory [33], the rate constants of $\text{C}_3\text{H}_3 + \text{HO}_2 = \text{C}_3\text{H}_4\text{-A} + \text{O}_2$ (R627), $\text{C}_3\text{H}_3 + \text{HO}_2 = \text{C}_3\text{H}_4\text{-P} + \text{O}_2$ (R628), $\text{C}_3\text{H}_4\text{-P} + \text{OH} = \text{C}_3\text{H}_3 + \text{H}_2\text{O}$ (R648), and $\text{C}_3\text{H}_4\text{-A} + \text{OH} = \text{C}_3\text{H}_3 + \text{H}_2\text{O}$ (R649) were derived from the analogy results in Ref. [38] in 2014.

The detailed HUST pyrrole model containing 1533 reactions and 249 species was provided in the supplemental material and compared to the Lumbieras model [11] in this study. Lumbieras model [11] consists of the pyrrole sub-mechanism, C₁-C₂ sub-mechanism [34], and benzene sub-mechanism [35]. Both of the kinetic models for pyridine oxidation proposed by Alzueta et al. [21] in 2002 and

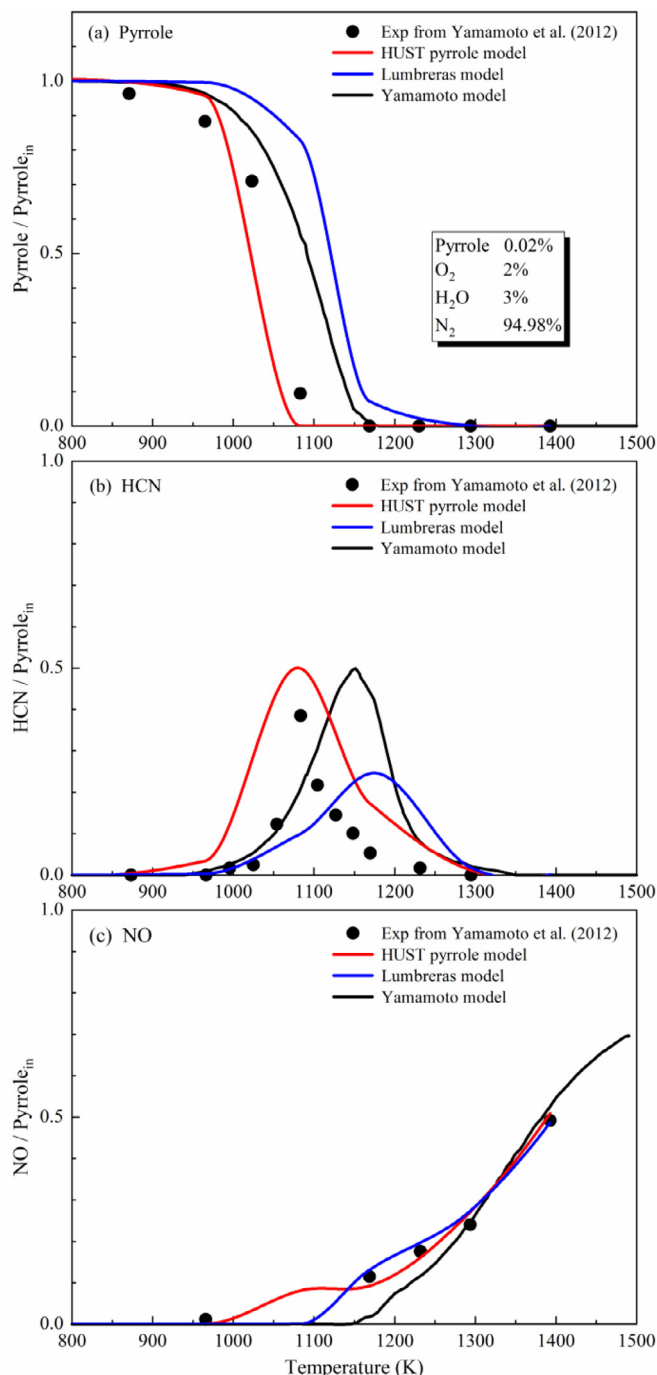


Fig. 6. Validation against species profiles of pyrrole oxidation from Yamamoto et al. [14] obtained under atmospheric pressure, the residence time is 251/T s.

Table 3Updated and modified reactions in this study. (The reaction rate coefficients are expressed as $k = AT^\beta \exp(-E_a/RT)$, units are $\text{cm}^3, \text{s}^{-1}$, and cal/mole).

No.	Reaction	A	β	E_a	Source
Pyrrole related reactions and their subsequent reactions					
R1237	PYRROLE=PYRLEYL+H	1.0E14	0	84,000	[9]
	Updated	3.0E15	0	107,000	[30]
R1238	PYRROLE+H<=>PYRLEYL+H ₂	2.0E13	0	5000	[9]
	Updated	2.0E14	0	5000	10*[9]
R1240	PYRROLE+OH=PYRLEYL+H ₂ O	3.0E06	0	-1310	[11]
	Updated	1.72E6	2.65	3646	[18]
R1257	PYRLNE+OH=PYRLEYL+H ₂ O	3.0E06	0	-1310	[11]
	Updated	1.72E6	2.65	3646	analogy to PYRROLE
R1251	PYRLNE=ALLYLCN	5.0E15	0	68,000	[9]
	Updated	5.0E16	0	68,000	10*[9]
R1531	PYRROLE<=>HNCPPROP	3.1E15	0	77,147	[13]
	Updated	3.1E16	0	77,147	10*[13]
R1273	AC ₃ H ₄ CN+H ₂ =>ALLYLCN+H	1.3E13	0	23,900	[9]
	Updated	1.3E14	0	23,900	10*[9]
R1284	AC ₃ H ₄ CN + H = ALLYLCN	3.0E13	0	0	[9]
	Updated	5.0E12	0	0	[31]
R1285	AC ₃ H ₄ CN + H = C ₃ H ₅ CN	3.0E13	0	0	[9]
	Updated	5.0E12	0	0	[31]
allene/propyne/propargyl radical related reactions updated					
R627	C ₃ H ₃ +HO ₂ <=>C ₃ H ₄ -A + O ₂	3.0E11	0	0	[36]
	Updated /reverse/	4.0E13	0	41,320	[37]
R628	C ₃ H ₃ +HO ₂ <=>C ₃ H ₄ -P + O ₂	2.5E12	0	0	[36]
	Updated /reverse/	3.0E13	0	42,630	[37]
R665	C ₃ H ₃ +O ₂ <=>CH ₂ CO+HCO	3.0E10	0	2868	[36]
	Updated	1.7E5	1.7	1500	[33]
R648	C ₃ H ₄ -P+OH<=>C ₃ H ₃ +H ₂ O	1.0E6	2	100	[36]
	Updated	4.94E6	2.027	1059.6	[37]
R649	C ₃ H ₄ -A+OH<=>C ₃ H ₃ +H ₂ O	5.3E6	2	2000	[36]
	Updated	1.482E6	2.492	1807.2	[37]
C ₀ -C ₁ related reactions updated					
R19	OH+HO ₂ =H ₂ O+O ₂	2.89E13	0	-500	[38]
	Updated	4.5E14	0	10,920	[39]
	Duplicate	7.0E12	0	-1092	[39]
R36	HCO+M = H+CO+M	5.7E11	0.66	14,870	[40]
	Updated CO ₂ third body efficiency /3.6/				[41]
R38	HCO+O ₂ =CO+HO ₂	7.58E12	0	410	[42]
	Updated	6.02E10	0.521	-521	[43]
R150	CH ₃ +HO ₂ =CH ₃ O+OH	1.0E12	0.269	-687.5	[44]
	Updated	1.04E13	0	-590	[45]
R151	CH ₃ +HO ₂ =CH ₄ +O ₂	1.16E5	2.230	-3022	[44]
	Updated	1.27E5	2.228	-3022	[45]

Wu et al. [22] in 2019 employed the pyrrole sub-mechanism from the Lumberras model [11].

4. Results and discussion

4.1. Experimental facilities and methods validation

The IDTs of pyrrole were measured under the condition of 1% pyrrole, 6.25% O₂, and 92.75% Ar at $P = 2.2$ bar, which is consistent with that in MacNamara et al. [12]. Figure 2 shows the comparison between the IDTs of pyrrole measured in the present study and those from MacNamara et al. [12]. From Fig. 2, our facilities and methods are reliable to carry out experiments.

4.2. Experimental results

Figure 3 presents the measured IDTs of pyrrole mixtures in the O₂/CO₂ atmosphere at various equivalence ratios at 5.2 bar. From Fig. 3, the pyrrole IDTs decrease with decreasing equivalence ratio. As shown in Fig. 4, the IDTs of pyrrole in the O₂/CO₂ and O₂/Ar atmospheres are compared. The results show that the pyrrole mixtures in O₂/CO₂ atmospheres have longer IDTs than those in O₂/Ar atmospheres.

4.3. Model validation and comparison

Figure 5 presents comparisons of pyrrole IDTs between experiments and the prediction results obtained by the HUST pyrrole model (red line), Lumberras model (black line), and Wu model

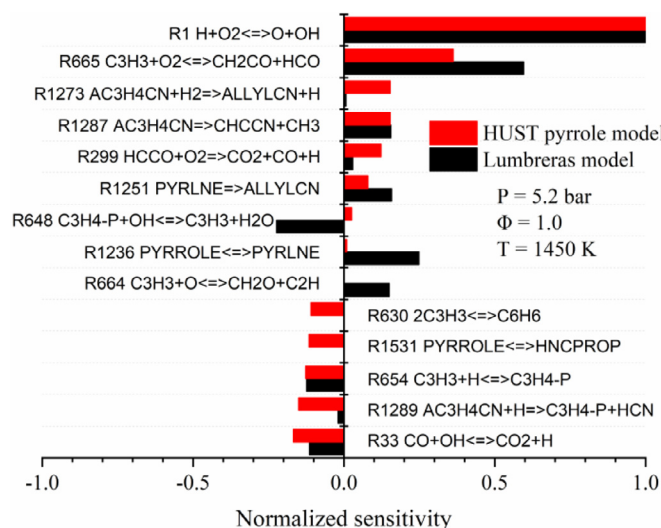


Fig. 7. Sensitivity analysis for the stoichiometric pyrrole mixture at 1450 K and 5.2 bar based on the HUST pyrrole model and Lumberras model.

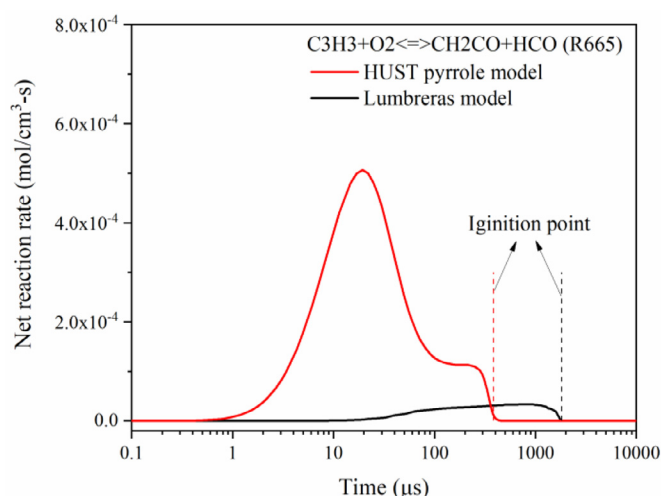


Fig. 8. Comparison between the net reaction rates of $C_3H_3 + O_2 = CH_2CO + HCO$ (R665) in HUST pyrrole model and those in Lumbreras model for the stoichiometric pyrrole mixture at 1450 K and 5.2 bar.

(blue line) in O_2/CO_2 and O_2/Ar atmospheres. As observed, the Lumbreras model and Wu model extremely overpredict the IDTs of all mixtures. HUST pyrrole model gives a satisfactory prediction of all the experimental data.

The species profiles are another key parameter to validate the reaction kinetic model. Figure 6 shows the predicted results of the three models for the profiles of pyrrole, HCN, and NO measured in a PFR at atmospheric pressure by Yamamoto et al. [14]. Red, black, and blue lines denote the HUST pyrrole model, Yamamoto model, and Lumbreras model, respectively. The experimental and modeling results of pyrrole consumption under the condition of 3 vol.% H_2O addition are shown in Fig. 6(a), the HUST pyrrole model shows the best performance for fuel consumption, whereas both the Yamamoto model and Lumbreras model underestimate the reactivity of pyrrole. Figure 6(b) shows the profile of HCN formation. The HUST pyrrole model provides better simulation results than the Yamamoto model and Lumbreras model, especially for the peak temperature and the maximum concentration of HCN. Figure 6(c) provides the mole fraction of NO. The HUST pyrrole model captures well the profile of NO formation. The Lumbreras

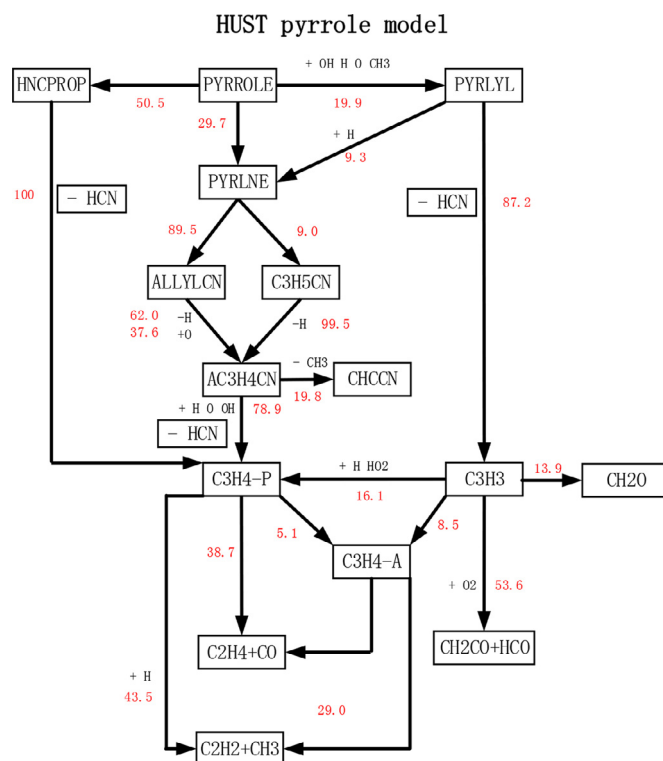


Fig. 10. Flux analysis for the stoichiometric pyrrole mixture at 1450 K and 5.2 bar based on the HUST pyrrole model.

model overpredicts the onset temperature while the Yamamoto model significantly underestimates the NO formation. More validations of the HUST pyrrole model against the experimental data of pyrrole from jet-stirred reactor (JSR) [46], shock tube [9], and PFR [11,14] are provided in the supplemental material. The validations of the HUST pyrrole model against the pyridine related experiments are also provided in the supplemental material.

Therefore, the HUST pyrrole model is reasonable and reliable.

The HUST pyrrole model predicts well the IDTs of pyrrole at the pressure of 5.2 bar and $\Phi = 1.0$, while the Lumbreras model significantly overestimates the IDTs. A temperature sensitivity analysis for the mixture 2 ($P = 5.2$ bar, $\Phi = 1.0$) at the ignition

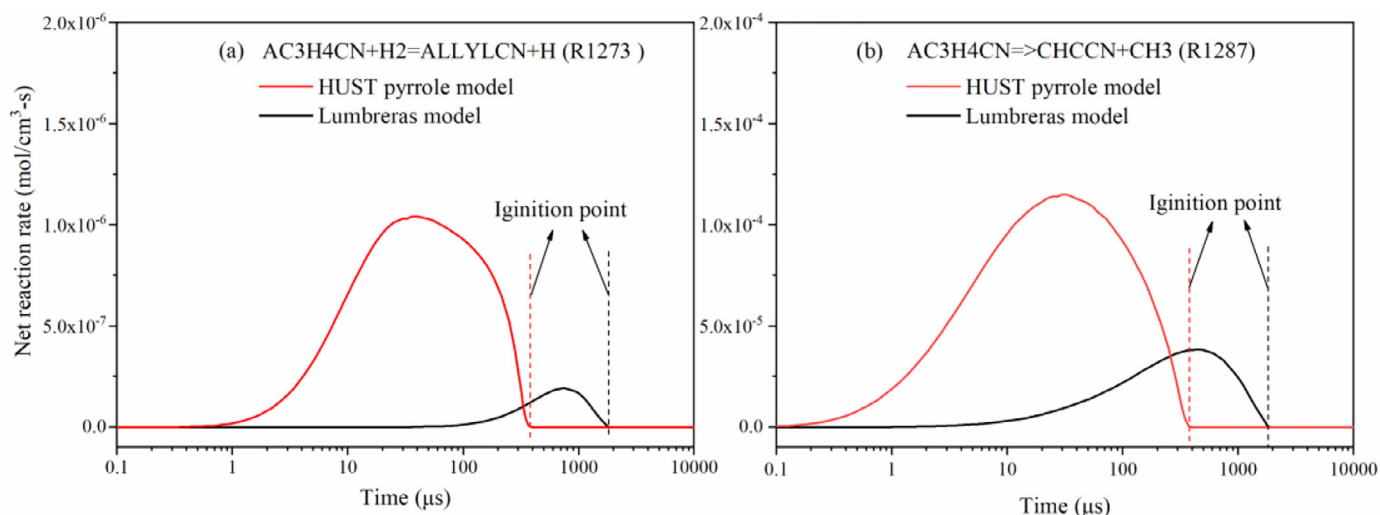


Fig. 9. Comparison between the net reaction rates in HUST pyrrole model and those in Lumbreras model for (a) $AC_3H_4CN + H_2 = ALLYLCN + H$ (R1273) and (b) $AC_3H_4CN = CHCCN + CH_3$ (R1287) for the stoichiometric pyrrole mixture at 1450 K and 5.2 bar.

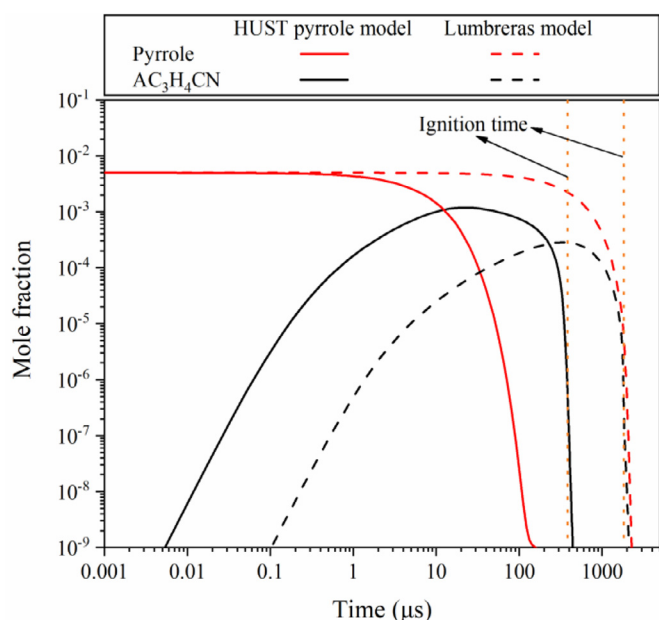


Fig. 11. Selected species history in the ignition of the stoichiometric pyrrole mixture at 1450 K and 5.2 bar.

point was carried out at 1450 K based on the HUST pyrrole model and Lumbrellas model, as shown in Fig. 7. Positive values indicate an ignition-promoting effect, while negative values indicate an ignition-inhibiting effect. The normalized result of the sensitivity analysis demonstrates that R665 has the 2nd positive sensitivity coefficients in both the HUST pyrrole model and Lumbrellas model, indicating that R665 is a crucial reaction to promote the pyrrole ignition. Figure 8 shows the net reaction rates of R665 in the HUST pyrrole model and Lumbrellas model at $T = 1450$ K and $P = 5.2$ bar. The rate constants of R665 in the HUST pyrrole model were derived from the theoretical calculation results in Ref. [33], whereas it was sourced from Laskin et al. [36] in the Lumbrellas model. As observed in Fig. 8, the net reaction rate of R665 in the HUST pyrrole model is much higher than that in the Lumbrellas model. Consequently, the underestimation of R665 in the Lumbrellas model is the main reason for the significant overprediction of the IDTs of pyrrole.

Two cyanoallyl radical (AC_3H_4CN) related reactions R1273 and R1287 ($AC_3H_4CN \Rightarrow CHCCN + CH_3$) rank 3rd and 4th in the promoting-ignition reactions in the HUST pyrrole model. Figure 9(a) and (b) displays the net reaction rates of R1273 and R1287 in the HUST pyrrole model and Lumbrellas model at $T = 1450$ K and $P = 5.2$ bar. The net reaction rates of R1273 and R1287 in the HUST pyrrole model reach the peak faster than those in Lumbrellas model, which also leads to the overprediction of the IDTs in the Lumbrellas model.

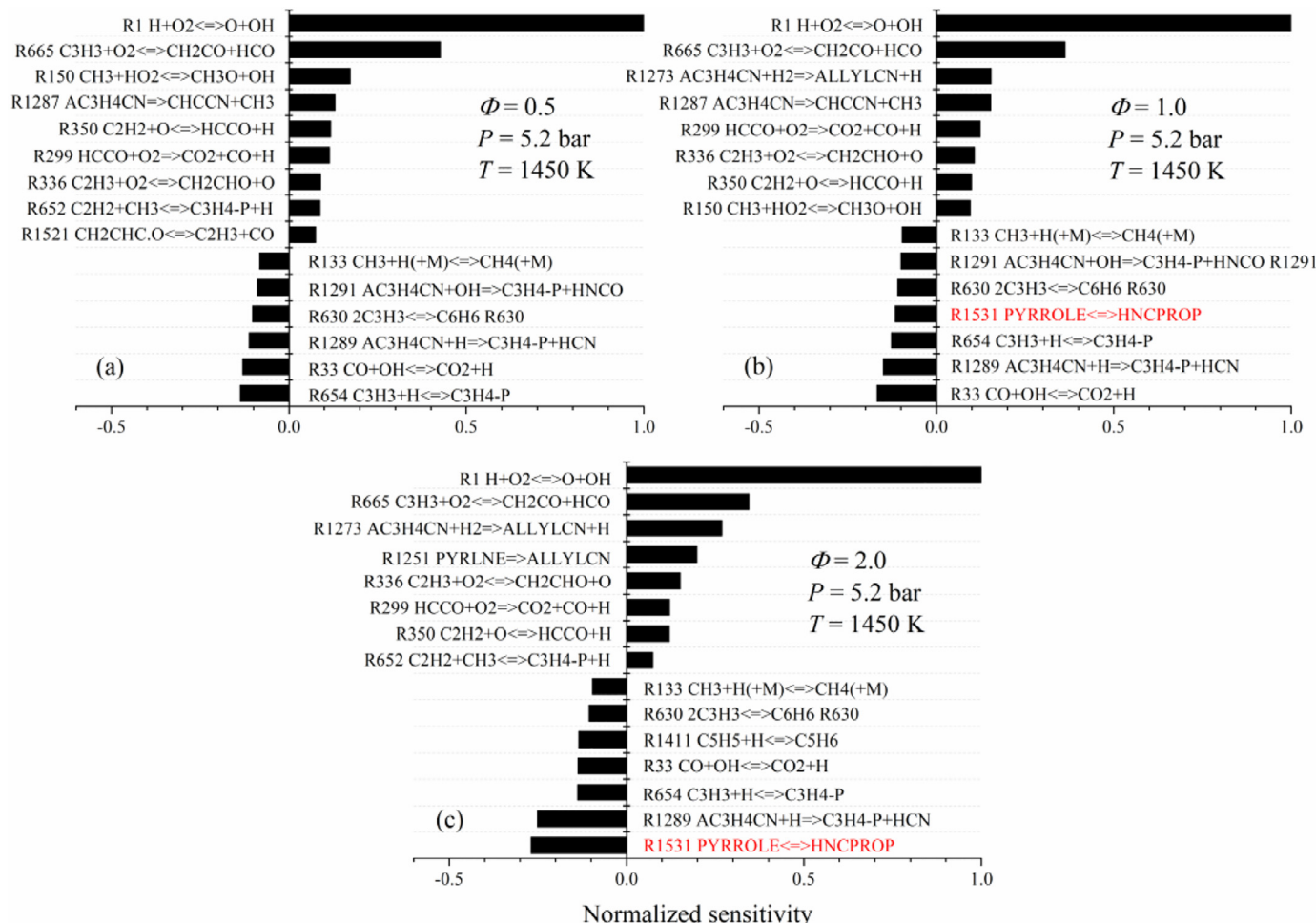


Fig. 12. Sensitivity analysis using the HUST pyrrole model at $P = 5.2$ bar and $T = 1450$ K for different equivalence ratios.

Updating the rate constants of R1273 in the HUST pyrrole model is an important reason for the higher net reaction rates of R1273. From Fig. 7, R1251 (PYRLNE = ALLYLCN), which were enhanced in this study, and R1236 (PYRROLE = PYRLNE) are sensitive to promoting the ignition in both HUST pyrrole model and Lumbreras model. The flux analyses were performed using the HUST pyrrole model and Lumbreras model at 20% fuel consumption point, as shown in Fig. 10 and Fig. S10, respectively. To avoid clutter, only fluxes higher than 5% are shown. As shown in Fig. 10 and Fig. S10, the path $\text{PYRROLE} \rightarrow \text{PYRLNE} \rightarrow \text{ALLYLCN} \rightarrow \text{AC}_3\text{H}_4\text{CN}$ is the main channel to produce $\text{AC}_3\text{H}_4\text{CN}$. Figure 11 illustrates the temporal profiles of pyrrole and $\text{AC}_3\text{H}_4\text{CN}$. The mole fraction of $\text{AC}_3\text{H}_4\text{CN}$ in the HUST pyrrole model is higher than that in the Lumbreras model. Therefore, the underestimation of R1273 and 1251 in the Lumbreras model is the reason for the overprediction of the IDTs.

The path $\text{PYRROLE} \rightarrow \text{PYRLYL} \rightarrow \text{HCN}$, one of the main channels to produce HCN, is enhanced in the HUST pyrrole model due to the larger rate constants of R1238 ($\text{PYRROLE} + \text{H} = \text{PYRLYL} + \text{H}_2$) and R1240 ($\text{PYRROLE} + \text{OH} = \text{PYRLYL} + \text{H}_2\text{O}$) compared to those in the Lumbreras model. Moreover, the path $\text{PYRROLE} \rightarrow \text{HNCPPROP}$, which accounts for 50.5% in the HUST pyrrole model, is absent in the Lumbreras model, as shown in Fig. 10 and Fig. S10. Thus, the path $\text{PYRROLE} \rightarrow \text{HNCPPROP} \rightarrow \text{HCN}$ in HUST pyrrole model, one of the main channels to produce HCN, also produces much more HCN than the direct decomposition path $\text{PYRROLE} \rightarrow \text{HCN}$ in Lum-

breras model. Therefore, the absence of HNCPPROP and the underestimation of R1238, and R1240 in Lumbreras model leads to the underprediction of the peak concentration of HCN and the overprediction of the temperature of the HCN maximum.

Obviously, the addition of HNCPPROP and modification of R665, R1273, R1251, R1238, and R1240 in HUST pyrrole model, as listed in Table 3, significantly improve the prediction of the pyrrole oxidation.

As can be seen from Fig. 3, the pyrrole IDTs increase with the equivalence ratio at = 5.2 bar. The temperature sensitivity analyses using the HUST pyrrole model were performed at $\Phi = 0.5, 1.0$, and 2.0, as shown in Fig. 12. As observed, the negative sensitivity of R1531 significantly increases as the equivalence ratio increases from 0.5 to 1.0 and 2.0. The R1531 inhibits the ignition of pyrrole due to the competition with the ignition enhancing channels $\text{PYRROLE} \rightarrow \text{PYRLNE}$ and $\text{PYRROLE} \rightarrow \text{PYRLYL}$ for pyrrole. The increase in the concentration of pyrrole under fuel-rich conditions leads to the enhancement of the ignition-inhibiting reaction R1531. Therefore, the equivalence ratio influences the IDTs of pyrrole mainly through R1531.

4.4. The effect of CO_2 on the pyrrole ignition

From Fig. 5(c), the IDTs of pyrrole in O_2/CO_2 atmospheres are longer than those in O_2/Ar atmospheres. To isolate the physical ef-

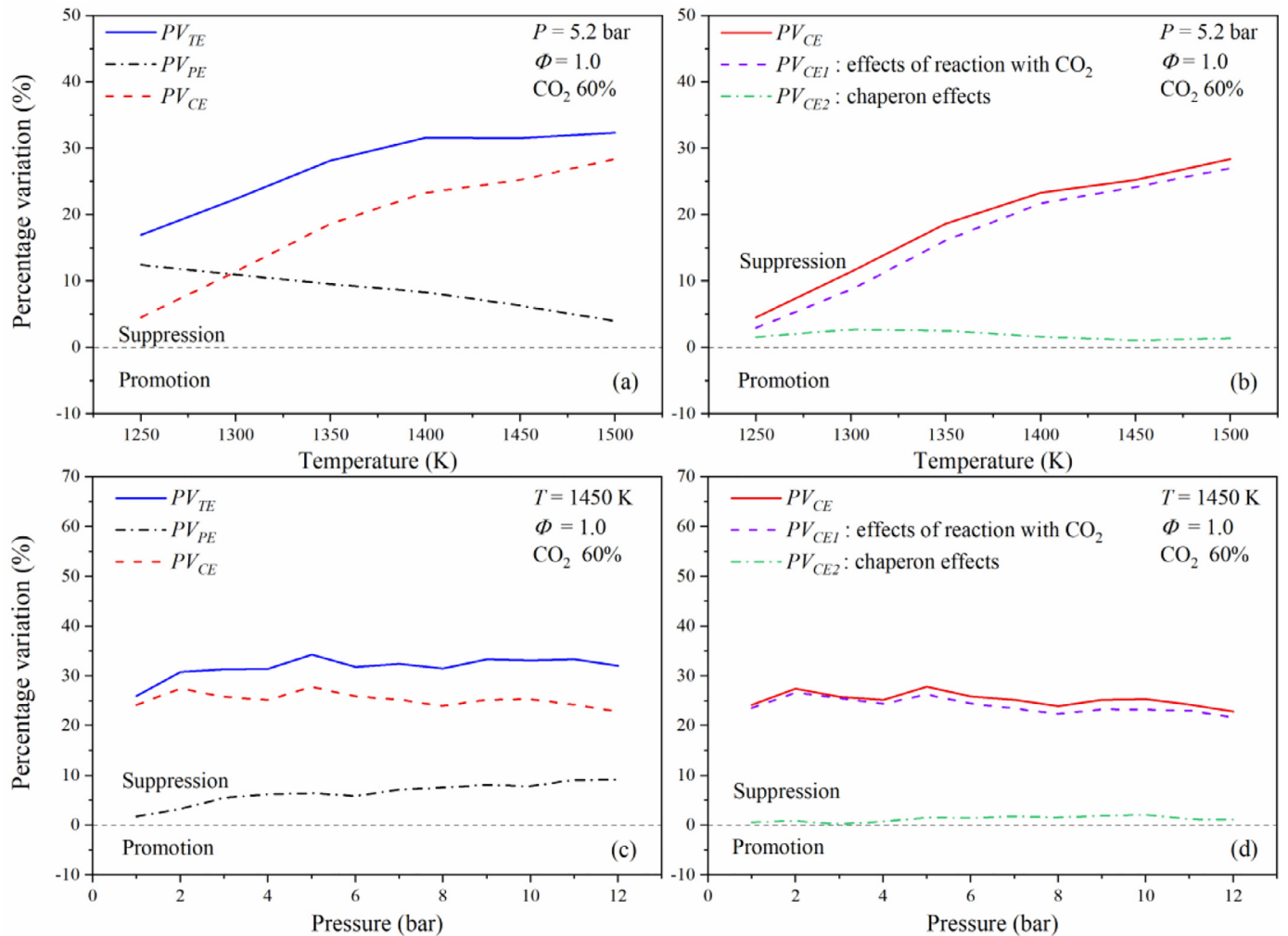


Fig. 13. The chemical effect (CE) and physical effect (PE) of CO_2 on the IDTs of pyrrole.

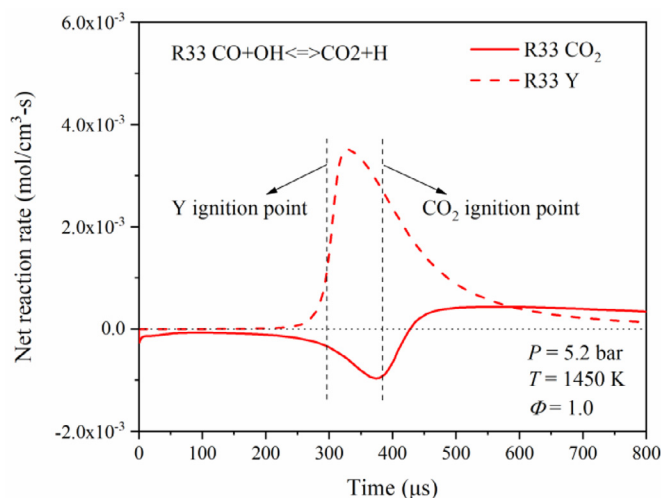


Fig. 14. Comparison of the net reaction rates between CO₂ and Y atmospheres for R33 at 5.2 bar and 1450 K.

fect (PE) and chemical effect (CE) of CO₂ on the IDTs of pyrrole, species X, which has the same thermochemical properties as CO₂ and is chemically inert, was introduced. The percentage variation (PV) [47] of the pyrrole IDTs [47] was introduced to quantitatively reveal the PE and CE of CO₂ and defined as:

$$PV_{PE} = (\tau_X - \tau_{N_2}) / \tau_X \quad (3)$$

$$PV_{CE} = (\tau_{CO_2} - \tau_X) / \tau_X \quad (4)$$

The sum of PV_{PE} and PV_{CE} demonstrates the total effect of CO₂ (PV_{TE}).

The CE of CO₂ on the pyrrole ignition can be further divided into the effect of the reactions with CO₂ (termed CE1) and the chaperon effect (termed CE2). To separate the CE1 and CE2 of CO₂, species Y, which has the same thermochemical properties and third-body collision efficiencies as CO₂ and do not participate in any chemical reactions, was introduced. Consequently, the CE1 can be denoted as the pyrrole IDTs deviation between the O₂/Y atmosphere and O₂/X atmosphere, and the CE2 can be expressed as the pyrrole IDTs deviation between the O₂/CO₂ atmosphere and O₂/Y atmosphere. The PV of the CE1 and CE2 was defined as:

$$PV_{CE1} = (\tau_{CO_2} - \tau_Y) / \tau_X \quad (5)$$

$$PV_{CE2} = (\tau_Y - \tau_X) / \tau_X \quad (6)$$

Positive PV values indicate an ignition-suppressing effect, while negative PV values indicate an ignition-promoting effect. Figure 13(a) and (b) show the PV profile as a function of the temperature at 5.2 bar, $\Phi = 1.0$. From Fig. 13(a), the PV_{TE} value increases gradually with temperature and then remains nearly unchanged above 1400 K. The PV_{PE} is always a positive value and decreases as the temperature increases. The PV_{CE} is always a positive value as well and increases with the temperature. The PV_{PE} and PV_{CE} have an intersection at nearly 1300 K, indicating that the PE of CO₂ is the main effect at lower temperatures and the CE of CO₂ dominates at $T > 1300$ K. Furthermore, as observed in Fig. 13(b), the PV_{CE1} is consistently predominant in the PV_{CE} and the PV_{CE2} value hovers around 1%. We can attribute the inhibition effect of CO₂ to the reactions with CO₂. Figure 13(c) and (d) show the PV profile as a function of pressure for the stoichiometric pyrrole mixture at 1450 K. As shown in Fig. 13(c), the PV_{TE} value increases from 25.9% to 30.7% as the pressure increases from 1 to 2 bar and then remains nearly unchanged at 2–12 bar. In other

words, the total inhibition effect of CO₂ is approximately independent of the pressure at $P > 2$ bar. In addition, the CE of CO₂ (PV_{CE}) is much stronger than the PE of CO₂ (PV_{PE}). From Fig. 13(d), the CE of CO₂ (PV_{CE}) is mainly caused by the CE1 of CO₂ (PV_{CE1}) and the CE2 of CO₂ can be almost ignored. According to the sensitivity analysis under CO₂ atmospheres, as shown in Fig. 7, R33 (CO + OH = CO₂ + H) is the most inhibition reaction for the pyrrole ignition. The net reaction rates of R33 in CO₂ and Y atmospheres are shown in Fig. 14. R33 proceeds in the reverse direction in CO₂ atmospheres due to the high concentration of CO₂, competing with the chain branching reaction R1 (H + O₂ = O + OH) for H radicals and thus leading to the inhibition of the pyrrole ignition. Obviously, the CE1 is sourced from R33.

5. Conclusions

The auto-ignition properties of the pyrrole mixture in the O₂/CO₂ atmosphere were investigated at an elevated pressure of 5.2 bar, temperatures from 1271 to 1645 K, and equivalence ratios $\Phi = 0.5, 1.0$, and 2.0 in a shock tube. The results indicated that the equivalence ratio has important effects on the IDTs of pyrrole. The IDTs of pyrrole in O₂/CO₂ atmospheres are longer than those in O₂/Ar atmospheres.

A pyrrole oxidation model (HUST pyrrole model) has been developed by updating 19 reactions on the basis of our previous work. HUST pyrrole model gives a satisfactory prediction of the pyrrole IDTs in the O₂/Ar atmosphere (measured by MacNamara et al.) and O₂/CO₂ atmosphere (measured in this study) and the profiles of pyrrole, HCN, and NO measured by Yamamoto et al. in a PFR. The addition of HNCPROP and modification of R665, R1273, R1251, R1238, and R1240 in the HUST pyrrole model significantly improve the prediction performance for the pyrrole oxidation.

R665, R1251, and R1273 play a significant role to promote the pyrrole ignition. The path PYRROLE → PYRLNE → AL-LYLCN → AC₃H₄CN is the main pathway to produce AC₃H₄CN, which is the main source for R1273 and R1287, and the concentration of AC₃H₄CN is underestimated in Lumbreras model due to the underestimation of R1251. Therefore, the underestimation of R665, R1251, R1273 in the Lumbreras model results in the significant overprediction of the IDTs of pyrrole.

The main paths of the production of HCN in HUST pyrrole model are (1) the path PYRROLE → HNCPROP → HCN, and (2) the path of PYRLYL to HCN via R1243 (PYRLYL = C₃H₃ + HCN), in which R1238 (PYRROLE + H = PYRLYL + H₂) and R1240 (PYRROLE + OH = PYRLYL + H₂O) provide PYRLYL to R1243. Therefore, the absence of HNCPROP and the underestimation of R1238 and R1240 in the Lumbreras model lead to the underprediction of the peak concentration of HCN and the overprediction of the temperature of the HCN maximum.

The overall effect of CO₂ suppresses the pyrrole ignition consistently. The chemical effect of CO₂ plays a dominant role at $T > 1300$ K, in which effects of the reactions with CO₂ dominates. The effect of the reactions with CO₂ can be attributed to R33 (CO + OH = CO₂ + H). The influence of CO₂ on the pyrrole ignition is approximately independent of the pressure at pressures more than 2 bar.

Declaration of Competing Interest

None

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2021.111458](https://doi.org/10.1016/j.combustflame.2021.111458).

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